

Experiment 08:

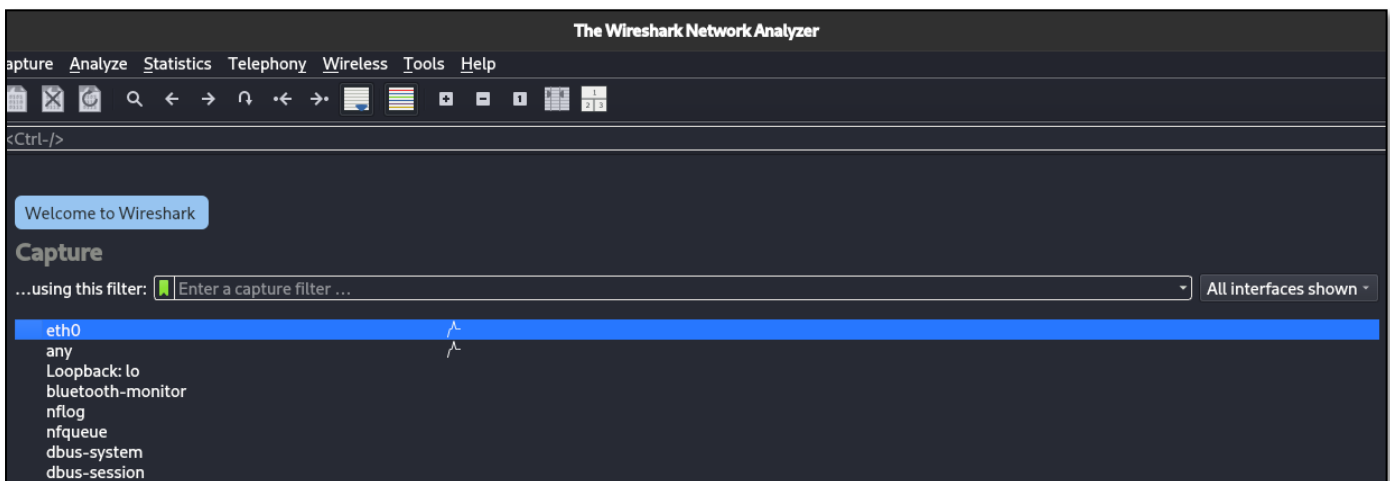
Step 1: Launch Wireshark



Observation:

Wireshark was located in the applications dock and launched successfully.

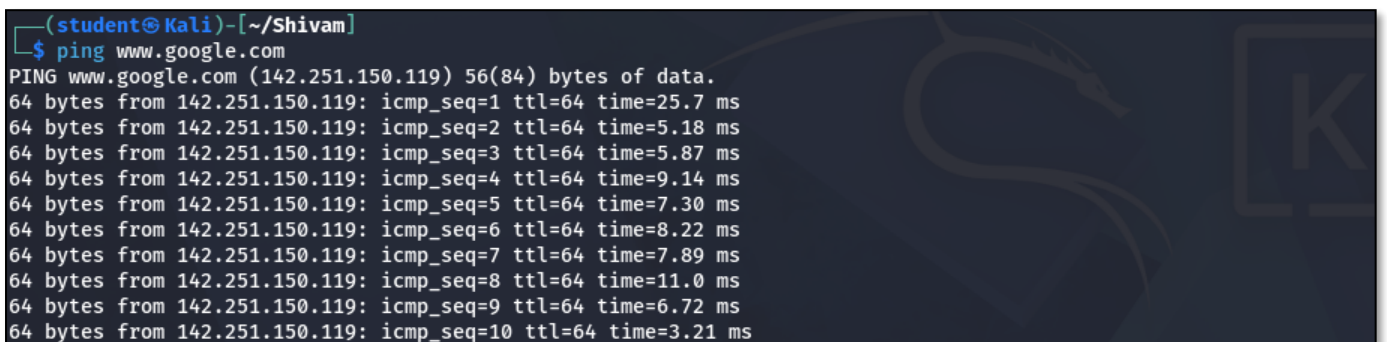
Step 2: Select Network Interface



Observation:

The available network interfaces were displayed, and the eth0 interface was selected for packet capture.

Step 3: Start Packet Capture



Capturing from eth0									
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help									
Apply a display filter ... <Ctrl-F>									
No.	Time	Source	Destination	Protocol	Length	Info			
2	0.004672734	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=109/27904,	ttl=64 (request in 1)
3	1.002718715	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=110/28160,	ttl=64 (reply in 4)
4	1.006715062	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=110/28160,	ttl=64 (request in 3)
5	2.005646767	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=111/28416,	ttl=64 (reply in 6)
6	2.011127130	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=111/28416,	ttl=64 (request in 5)
7	3.007544370	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=112/28672,	ttl=64 (reply in 8)
8	3.022952759	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=112/28672,	ttl=64 (request in 7)
9	4.009952982	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=113/28928,	ttl=64 (reply in 10)
10	4.035601945	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=113/28928,	ttl=64 (request in 9)
11	5.012003340	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=114/29184,	ttl=64 (reply in 12)
12	5.088137186	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=114/29184,	ttl=64 (request in 11)
13	6.015047143	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=115/29440,	ttl=64 (reply in 14)
14	6.022560359	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=115/29440,	ttl=64 (request in 13)
15	7.016710734	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=116/29696,	ttl=64 (reply in 16)
16	7.021639571	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=116/29696,	ttl=64 (request in 15)

Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0	0000	52	54	00	12	35	00	08	00	27	75	12	d2	08	00	45	00
Ethernet II, Src: PCSSystemtec_75:12:d2 (08:00:27:75:12:d2), Dst: 52:54:00:12:35:00 (52:54:00:12:35:00)	0010	00	54	09	24	40	00	40	01	00	04	0a	00	02	0f	8e	fb
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 142.251.150.119	0020	96	77	08	00	fe	1b	5d	28	00	6d	23	85	c8	6a	00	00
Internet Control Message Protocol	0030	00	00	eb	8b	06	00	00	00	00	00	10	11	12	13	14	15
	0040	16	17	18	19	1a	1b	1c	1d	1e	1f	20	21	22	23	24	25
	0050	26	27	28	29	2a	2b	2c	2d	2e	2f	30	31	32	33	34	35
	0060	36	37														

Observation:

- The ping www.google.com command was executed, and ICMP echo replies were received successfully.
- Wireshark captured and displayed ICMP echo requests and replies between the source and destination IP addresses.

Step 4: ICMP Packet Analysis

icmp									
No.	Time	Source	Destination	Protocol	Length	Info			
92	44.150941371	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=109/27904,	ttl=64 (request in 1)
93	45.144812920	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=110/28160,	ttl=64 (reply in 4)
94	45.150714539	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=110/28160,	ttl=64 (request in 3)
95	46.146125078	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=111/28416,	ttl=64 (reply in 6)
96	46.158444424	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=111/28416,	ttl=64 (request in 5)
97	47.151410017	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=112/28672,	ttl=64 (reply in 8)
98	47.158051535	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=112/28672,	ttl=64 (request in 7)
99	48.158683333	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=113/28928,	ttl=64 (reply in 10)
100	48.164373405	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=113/28928,	ttl=64 (request in 9)
101	49.164348657	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=114/29184,	ttl=64 (reply in 12)
102	49.171466795	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=114/29184,	ttl=64 (request in 11)
103	50.168807334	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=115/29440,	ttl=64 (reply in 14)
104	50.174093449	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=115/29440,	ttl=64 (request in 13)
105	51.227671538	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=116/29696,	ttl=64 (reply in 16)
106	51.233403133	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=116/29696,	ttl=64 (request in 15)
107	52.233216310	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=117/30000,	ttl=64 (reply in 18)
108	52.238072757	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=117/30000,	ttl=64 (request in 17)
109	53.238908291	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=118/30304,	ttl=64 (reply in 20)
110	53.245716382	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=118/30304,	ttl=64 (request in 19)
113	54.242835152	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=119/30608,	ttl=64 (reply in 22)
114	54.255022939	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=119/30608,	ttl=64 (request in 21)
115	55.244477534	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=120/30912,	ttl=64 (reply in 24)
116	55.248671822	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=120/30912,	ttl=64 (request in 23)
117	56.245794125	10.0.2.15	142.251.150.119	ICMP	98	Echo (ping)	request	id=0x5d28, seq=121/31216,	ttl=64 (reply in 26)
118	56.250144045	142.251.150.119	10.0.2.15	ICMP	98	Echo (ping)	reply	id=0x5d28, seq=121/31216,	ttl=64 (request in 25)

Observation:

ICMP packets were filtered and analyzed, showing echo requests and replies with their source and destination IP addresses.

Step 5: ARP Packet Analysis

No.	Time	Source	Destination	Protocol	Length	Info
31	14.093376771	PCSSystemtec_75:12:...	52:54:00:12:35:00	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
32	14.093933569	52:54:00:12:35:00	PCSSystemtec_75:12:...	ARP	64	10.0.2.2 is at 52:54:00:12:35:00
111	53.263715115	PCSSystemtec_75:12:...	52:54:00:12:35:00	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
112	53.264619946	52:54:00:12:35:00	PCSSystemtec_75:12:...	ARP	64	10.0.2.2 is at 52:54:00:12:35:00
193	92.436621179	PCSSystemtec_75:12:...	52:54:00:12:35:00	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
194	92.437249946	52:54:00:12:35:00	PCSSystemtec_75:12:...	ARP	64	10.0.2.2 is at 52:54:00:12:35:00

Observation:

ARP requests and replies were captured, showing IP-to-MAC address resolution on the network.

Step 6: DNS Packet Analysis

```
(student@Kali)-[~/Shivam]
$ nslookup google.com
Server:      192.168.7.254
Address:     192.168.7.254#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.174.101
Name:   google.com
Address: 192.178.174.138
Name:   google.com
Address: 192.178.174.102
Name:   google.com
Address: 192.178.174.139
Name:   google.com
Address: 192.178.174.113
Name:   google.com
Address: 192.178.174.100
Name:   google.com
Address: 2404:6800:4000:1010::8a
Name:   google.com
Address: 2404:6800:4000:1010::71
Name:   google.com
Address: 2404:6800:4000:1010::65
Name:   google.com
Address: 2404:6800:4000:1010::66
```

No.	Time	Source	Destination	Protocol	Length	Info
732	357.546277732	10.0.2.15	192.168.7.254	DNS	70	Standard query 0xba16 A google.com
733	357.555707252	192.168.7.254	10.0.2.15	DNS	166	Standard query response 0xba16 A google
734	357.556094284	10.0.2.15	192.168.7.254	DNS	70	Standard query 0x490c AAAA google.com
735	357.565934814	192.168.7.254	10.0.2.15	DNS	182	Standard query response 0x490c AAAA goo

Observation:

- The nslookup google.com command displayed the resolved IPv4 and IPv6 addresses of the domain.
- Wireshark displayed DNS queries and responses for the A and AAAA records of google.com.

Step 7: TCP Packet Analysis

```
(student@Kali)-[~/Shivam]
$ curl https://example.com
<!doctype html><html lang=en><head><meta charset=utf-8><link rel=icon href=data:,><me:light dark;background:light-dark(#eee,#222)}body{font:16px/1.6 system-ui,sans-ser
e in documentation examples without needing permission. This is not a service; avoid
```

No.	Time	Source	Destination	Protocol	Length	Info
896	433.948660685	fd17:625c:f037:2:eb...	2606:4700:10::ac42:...	TCP	94	39000 → 443 [SYN]
899	434.151537712	10.0.2.15	172.66.147.243	TCP	74	58070 → 443 [SYN]
900	434.152281381	2606:4700:10::ac42:...	fd17:625c:f037:2:eb...	TCP	74	443 → 39000 [RST,
901	434.154589668	172.66.147.243	10.0.2.15	TCP	60	443 → 58070 [SYN,
902	434.154696781	10.0.2.15	172.66.147.243	TCP	54	58070 → 443 [ACK]
903	434.158446458	10.0.2.15	172.66.147.243	TLSv1.3	1751	Client Hello (SNI=
904	434.159357291	172.66.147.243	10.0.2.15	TCP	60	443 → 58070 [ACK]
905	434.159357767	172.66.147.243	10.0.2.15	TCP	60	443 → 58070 [ACK]
906	434.196421270	172.66.147.243	10.0.2.15	TLSv1.3	1494	Server Hello, Char
907	434.196421653	172.66.147.243	10.0.2.15	TCP	1494	443 → 58070 [ACK]
908	434.196545285	10.0.2.15	172.66.147.243	TCP	54	58070 → 443 [ACK]
909	434.196789676	10.0.2.15	172.66.147.243	TCP	54	58070 → 443 [ACK]
910	434.280488150	172.66.147.243	10.0.2.15	TCP	1494	443 → 58070 [ACK]
911	434.280488736	172.66.147.243	10.0.2.15	TLSv1.3	813	Application Data
912	434.280596495	10.0.2.15	172.66.147.243	TCP	54	58070 → 443 [ACK]
913	434.280967893	10.0.2.15	172.66.147.243	TCP	54	58070 → 443 [ACK]
914	434.285498429	10.0.2.15	172.66.147.243	TLSv1.3	134	Change Cipher Spec
915	434.286058678	10.0.2.15	172.66.147.243	TLSv1.3	177	Application Data
916	434.286447500	172.66.147.243	10.0.2.15	TCP	60	443 → 58070 [ACK]
917	434.286447984	172.66.147.243	10.0.2.15	TCP	60	443 → 58070 [ACK]
918	434.286475562	10.0.2.15	172.66.147.243	TLSv1.3	102	Application Data
919	434.287564645	172.66.147.243	10.0.2.15	TCP	60	443 → 58070 [ACK]
920	434.343870982	172.66.147.243	10.0.2.15	TLSv1.3	1439	Application Data,
921	434.344064847	10.0.2.15	172.66.147.243	TLSv1.3	85	Application Data
922	434.344181388	10.0.2.15	172.66.147.243	TLSv1.3	102	Application Data

Observation:

- The curl https://example.com command retrieved the website's HTML content successfully over HTTPS.
- Wireshark displayed TCP connection establishment packets followed by TLS-encrypted communication.

Step 8: UDP Packet Analysis

No.	Time	Source	Destination	Protocol	Length	Info
187	91.260357399	10.0.2.15	192.46.215.60	NTP	90	NTP Version 4, client
188	91.273142583	192.46.215.60	10.0.2.15	NTP	90	NTP Version 4, server

Observation:

UDP packets were captured, showing NTP client and server communication for time synchronization.

Step 9: HTTP Packet Analysis

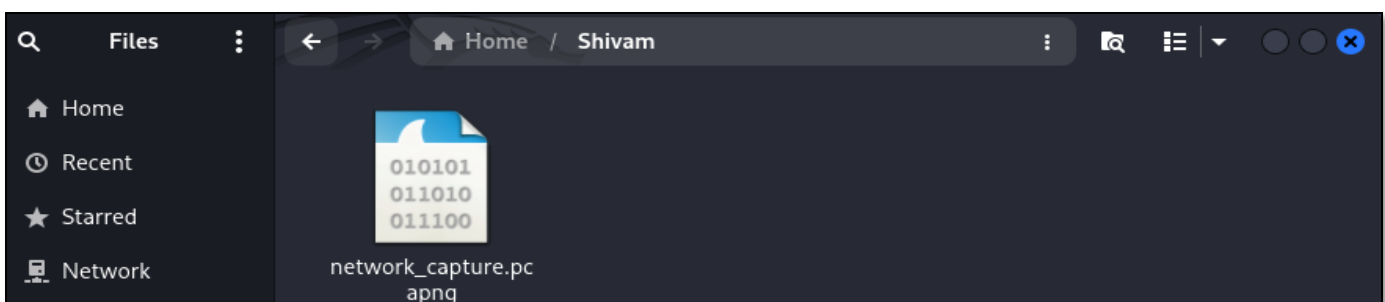
```
(student@Kali)-[~/Shivam]
$ curl http://neverssl.com
<html>
  <head>
    <title>NeverSSL - Connecting ... </title>
    <style>
      body {
        font-family: Montserrat, helvetica, arial, sans-serif;
        font-size: 16px;
        color: #444444;
        margin: 0;
      }
      h2 {
        font-weight: 700;
        font-size: 1.6em;
        margin-top: 30px;
      }
      p {
        line-height: 1.6em;
      }
      .container {
        max-width: 650px;
        margin: 20px auto 20px auto;
        padding-left: 15px;
        padding-right: 15px
      }
      .header {
        background-color: #42C0FD;
        color: #FFFFFF;
        padding: 10px 0 10px 0;
        font-size: 2.2em;
      }
      .notice {
```

http						
No.	Time	Source	Destination	Protocol	Length	Info
1111	518.194745823	10.0.2.15	34.223.124.45	HTTP	130	GET / HTTP/1.1
1125	520.900748326	34.223.124.45	10.0.2.15	HTTP	68	HTTP/1.1 200 OK (text/html)

Observation:

- The curl http://neverSSL.com command retrieved the website's HTML content using HTTP.
- Wireshark displayed an HTTP GET request and an HTTP/1.1 200 OK response.

Step 10: Save Packet Capture File



Observation:

The captured network packets were saved in the network_capture.pcapng file for future analysis.